

APEX INSTITUTE OF TECHNOLOGY & SCIENCE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Bachelor of Technology (B.Tech) — Curriculum & Regulations (2024–2028)

1. Program Overview and Academic Regulations

The B.Tech in Computer Science and Engineering (CSE) is a 4-year (8 semesters) undergraduate program designed in alignment with standard AICTE university guidelines. The program aims to equip students with strong foundational concepts in computing systems, algorithmic problem solving, software engineering, data science, and generative artificial intelligence.

2. Graduation and Credit Requirements

Category	Category Description	Required Credits
PC	Program Professional Core Courses	68
PE	Professional Elective Courses	18
OE	Open University Electives	12
BS	Basic Science Courses (Maths, Physics)	24
ES	Engineering Science Courses	18
HSMC	Humanities, Social Sciences & Management	10
PROJ	Capstone Major Projects & Internship	16
Total	Minimum Total Credits for Degree Completion	166 Credits

3. Academic Regulations and Attendance Policy

- **Minimum Attendance Requirement:** A student must maintain a minimum of 75% attendance in each course.
- **Passing Criteria:** A minimum of 40% aggregate marks in continuous internal evaluations and end-semester examinations.
- **Grading Scale:** Standard 10-point scale: O (10), A+ (9), A (8), B+ (7), B (6), C (5), F (0).

Department of Computer Science and Engineering — First Year Structure

Semester 1 Course Structure

Course Code	Course Name	Type	Prerequisite	Credits
CS101	Introduction to Programming with C	Core	None	4 Credits
MA101	Engineering Mathematics I (Calculus & Matrices)	Basic Science	None	4 Credits
PH101	Engineering Physics	Basic Science	None	3 Credits
EE101	Basic Electrical & Electronics Engineering	Engineering Sci	None	3 Credits
HS101	Technical English Communication	Humanities	None	2 Credits
CS101P	C Programming Laboratory	Lab	None	1.5 Credits
PH101P	Physics Laboratory	Lab	None	1.5 Credits

Total Credits in Semester 1: 19 Credits

Semester 2 Course Structure

Course Code	Course Name	Type	Prerequisite	Credits
CS102	Data Structures and Algorithms	Core	CS101 (Programming)	4 Credits
MA102	Engineering Mathematics II (Differential Equations)	Basic Science	MA101	4 Credits
CS103	Digital Logic & System Design	Core	None	3 Credits
CH101	Environmental Science & Chemistry	Basic Science	None	3 Credits
ME101	Engineering Graphics and Design	Engineering Sci	None	2 Credits
CS102P	Data Structures Laboratory	Lab	CS101	2 Credits

Total Credits in Semester 2: 18 Credits

Department of Computer Science and Engineering — Second Year Structure

Semester 3 Course Structure (Third Semester)

Course Code	Course Name	Type	Prerequisite	Credits
CS201	Discrete Mathematical Structures	Core	MA101	3 Credits
CS202	Object-Oriented Programming with Java	Core	CS101 (Programming in C)	4 Credits
CS301	Database Management Systems (DBMS)	Core	CS102 (Data Structures)	4 Credits
CS204	Computer Organization and Architecture	Core	CS103 (Digital Logic)	4 Credits
MA201	Linear Algebra and Numerical Methods	Basic Science	MA101	3 Credits
CS301P	DBMS and SQL Laboratory	Lab	CS102	1.5 Credits
CS202P	Java OOP Laboratory	Lab	CS101	1.5 Credits

Total Credits in Semester 3 (Third Semester): 21 Credits

Semester 4 Course Structure (Fourth Semester)

Course Code	Course Name	Type	Prerequisite	Credits
CS207	Operating Systems Principles	Core	CS102 & CS204	4 Credits
CS208	Design and Analysis of Algorithms	Core	CS102 (Data Structures)	4 Credits
CS209	Software Engineering Principles & Agile	Core	None	3 Credits
MA202	Probability and Statistics for Engineers	Basic Science	MA101	3 Credits
HS201	Universal Human Values and Ethics	Humanities	None	2 Credits
CS207P	Operating Systems & Linux Shell Lab	Lab	CS101	1.5 Credits
CS208P	Algorithms Implementation Lab	Lab	CS102	1.5 Credits

Total Credits in Semester 4 (Fourth Semester): 19 Credits (Overall Year 2 Credits: 40 Credits)

Department of Computer Science and Engineering — Third Year Structure

Semester 5 Course Structure & Professional Electives

Course Code	Course Name	Type	Prerequisite	Credits
CS302	Computer Networks and Protocols	Core	CS207 (Operating Systems)	4 Credits
CS303	Theory of Computation and Automata	Core	CS201 (Discrete Math)	3 Credits
CS305	Machine Learning	Core	MA202 (Prob & Stats) & CS102	4 Credits
PE-I	Professional Elective I (List below)	Professional Elective	As per elective	3 Credits
OE-I	Open Elective I (University-wide)	Open Elective	None	3 Credits
CS302P	Computer Networks & Packet Tracing Lab	Lab	CS207	1.5 Credits
CS305P	Machine Learning with Python Lab	Lab	CS102	1.5 Credits

Total Credits in Semester 5: 20 Credits

Professional Elective Basket I (Available in Semester 5)

Course Code	Elective Subject Name	Prerequisites	Credits
CS311	Data Science Fundamentals	CS101 (Programming) or Python	3 Credits
CS312	Cloud Computing Architecture & AWS	CS207 (Operating Systems)	3 Credits
CS313	Cyber Security and Network Defense	CS207 & CS302 (Networks)	3 Credits
CS314	Computer Graphics and Virtual Reality	CS102 & MA201 (Linear Algebra)	3 Credits

Semester 6 Course Structure

Course Code	Course Name	Type	Prerequisite	Credits
CS306	Compiler Design Principles	Core	CS303 (Theory of Computation)	4 Credits
CS307	Web Technologies and Full-Stack Frameworks	Core	CS202 (Java OOP)	3 Credits
PE-II	Professional Elective II (NLP / Deep Learning / IoT)	Professional Elective	CS305	3 Credits
PE-III	Professional Elective III (Distributed Systems / DevOps)	Professional Elective	CS207	3 Credits
OE-II	Open Elective II	Open Elective	None	3 Credits
CS308P	Mini Project and Design Thinking	Project	CS301 & CS209	2 Credits

Total Credits in Semester 6: 18 Credits

Department of Computer Science and Engineering — Final Year Structure

Semester 7 Course Structure

Course Code	Course Name	Type	Prerequisite	Credits
CS401	Big Data Analytics & Distributed Computing	Core	CS301 (DBMS) & CS311	3 Credits
PE-IV	Professional Elective IV (Generative AI / Robotics)	Professional Elective	CS305	3 Credits
PE-V	Professional Elective V (Quantum Computing / Blockchain)	Professional Elective	CS208	3 Credits
OE-III	Open Elective III: Financial Engineering	Open Elective	None	3 Credits
CS402	Capstone Major Project Phase I	Core Project	All prior core courses	5 Credits

Total Credits in Semester 7: 17 Credits

Semester 8 Course Structure (Final Semester)

Course Code	Course Name	Type	Prerequisite	Credits
CS403	Capstone Major Project Phase II / Industrial Internship	Core Project	CS402	10 Credits
PE-VI	Professional Elective VI (Self-Paced / MOOC Online)	Professional Elective	Department approval	3 Credits

Total Credits in Semester 8: 13 Credits

Grand Total Degree Credits: 166 Credits

Detailed Course Syllabi and Examination Regulations

Course Syllabus: CS301 — Database Management Systems (DBMS)

Course Code: CS301 | **Credits:** 4 Credits (L-T-P: 3-1-0) | **Semester:** Third Semester (Semester 3)

Course Type: Professional Core | **Prerequisites:** CS102 Data Structures and Algorithms.

Syllabus Modules:

- **Module 1 (Foundations):** Database System Concepts, 3-tier Schema Architecture, Data Independence, Entity-Relationship (ER) Modeling, Extended ER.
- **Module 2 (Relational Model & Algebra):** Relational Data Model, Integrity Constraints, Tuple Relational Calculus, Domain Relational Calculus, SQL DDL/DML/DCL.
- **Module 3 (Normalization):** Functional Dependencies, Armstrong's Axioms, Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF), Lossless Join, Dependency Preservation.
- **Module 4 (Transaction Processing):** ACID Properties, Concurrency Control, Two-Phase Locking (2PL), Timestamp Ordering, Deadlock detection and recovery.
- **Module 5 (Storage & Indexing):** File Organization, RAID, B-Trees, B+ Tree Indexing, Hashing.

Course Syllabus: CS305 — Machine Learning

Course Code: CS305 | **Credits:** 4 Credits (L-T-P: 3-1-0) | **Semester:** Fifth Semester (Semester 5)

Course Type: Professional Core | **Prerequisites:** MA202 (Probability and Statistics) and CS102 (Data Structures).

Syllabus Modules:

- **Module 1 (Supervised Learning):** Linear Regression, Cost Functions, Gradient Descent, Logistic Regression, Multi-class Classification, Regularization (L1/L2 Lasso & Ridge).
- **Module 2 (Tree-based & Non-Parametric):** Decision Trees (ID3, C4.5, CART), Information Gain, Random Forests, Gradient Boosted Trees (XGBoost), k-Nearest Neighbors (k-NN).
- **Module 3 (Kernel Methods):** Support Vector Machines (SVM), Hyperplane margin maximization, Kernel trick (RBF, Polynomial kernels).
- **Module 4 (Unsupervised Learning):** k-Means Clustering, Hierarchical Agglomerative Clustering, Principal Component Analysis (PCA), Dimensionality Reduction.
- **Module 5 (Neural Networks Introduction):** Multilayer Perceptrons (MLP), Backpropagation algorithm, Activation functions (ReLU, Sigmoid, Softmax), Overfitting and Dropout.

Course Syllabus: CS311 — Data Science Fundamentals (Elective)

Course Code: CS311 | **Credits:** 3 Credits (L-T-P: 3-0-0) | **Semester:** Fifth Semester (Semester 5 Elective)

Course Type: Professional Elective | **Prerequisites:** CS101 (Programming with C) or Python Programming Fundamentals.

Syllabus Modules: Exploratory Data Analysis (EDA), NumPy, Pandas DataFrame operations, Data Cleaning, Matplotlib & Seaborn data visualization, Hypothesis Testing, Feature Engineering, Introduction to Big Data pipelines.